ATOMS

FORMULA SHEET | JEE MAINS | Let's Play With Physics

1. USEFUL CONSTANTS

Quantity	Value
$\dfrac{e^2}{4\pi\varepsilon_0}$	1.44 eV·nm = 1.44 MeV·fm
Bohr radius a_0	0.529 Å
Ground state energy (H)	−13.6 eV
Rydberg constant R	1.097×10^7 m^{-1}
$1/R$	≈ 912 Å
hc	1240 eV·nm = 12400 eV·Å
v_1 (H)	2.18×10^6 m/s ≈ $c/137$
Bohr magneton μ_B	9.27×10^{-24} A·m^2

$$\lambda\,[\text{Å}] = \frac{12400}{\Delta E\,[\text{eV}]} \qquad Rhc = 13.6 \text{ eV}$$

2. RUTHERFORD α-SCATTERING

Observations: most α-particles pass undeflected; a few deflect by large angles; about 1 in 8000 deflect by more than 90°.

Distance of closest approach (head-on):

$$r_0 = \frac{1}{4\pi\varepsilon_0}\cdot\frac{2Ze^2}{K}, \qquad K = \tfrac{1}{2}mv^2$$

Impact parameter:

$$b = \frac{1}{4\pi\varepsilon_0}\cdot\frac{Ze^2\cot(\theta/2)}{K} = \frac{r_0}{2}\cot\frac{\theta}{2}$$

Number scattered: $N(\theta) \propto \dfrac{1}{\sin^4(\theta/2)}$

$b = 0 \Rightarrow \theta = 180°$ (head-on, rebound); large $b \Rightarrow$ small θ
5.5 MeV α on gold: $r_0 \approx 4 \times 10^{-14}$ m

⋆ Common Pitfall

- r_0 gives an **upper limit** on nuclear size
- At closest approach (head-on) the α is momentarily at rest: all KE → PE
- Charge of α is $2e$ — don't forget the factor 2 in r_0
- Rutherford model fails: accelerating electron should radiate and spiral in; predicts continuous spectrum

3. BOHR'S POSTULATES

1. **Stationary orbits** — no radiation
2. **Quantisation:** $mvr = \dfrac{nh}{2\pi}$, $n = 1, 2, 3, \ldots$
3. **Frequency condition:** $h\nu = E_{n_2} - E_{n_1}$

de Broglie view: $2\pi r_n = n\lambda$ (whole number of waves fit the orbit)

⋆ Common Pitfall

- Bohr model is valid only for **one-electron** systems: H, He^+, Li^{2+}, …
- Angular momentum is quantised, not energy directly

4. BOHR MODEL RESULTS (Z, n)

$$r_n = 0.529\,\frac{n^2}{Z}\ \text{Å}, \qquad v_n = 2.18 \times 10^6\,\frac{Z}{n}\ \text{m/s}$$

$$E_n = -13.6\,\frac{Z^2}{n^2}\ \text{eV}$$

$$K = -E, \qquad U = 2E, \qquad K = -\tfrac{U}{2}$$

Quantity	∝	Quantity	∝
r	n^2/Z	T (period)	n^3/Z^2
v, p	Z/n	f, ω	Z^2/n^3
E, K, U	Z^2/n^2	Current I	Z^2/n^3
L	n	B at nucleus	Z^3/n^5
Force, a	Z^3/n^4	μ (mag. mom.)	n

Orbital current: $I = \dfrac{ev}{2\pi r}$

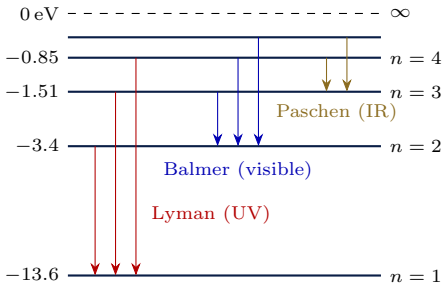
Magnetic moment: $\mu = \dfrac{nhe}{4\pi m} = n\mu_B$

$\dfrac{\mu}{L} = \dfrac{e}{2m}$ (gyromagnetic ratio, independent of n)

⋆ Common Pitfall

- Total energy is negative (bound); higher $n \Rightarrow$ less negative $\Rightarrow$ higher energy
- KE decreases with n but total energy increases
- $U = 2E$ assumes PE = 0 at infinity

5. ENERGY LEVELS & IONISATION



Ionisation energy (from n): $\dfrac{13.6\,Z^2}{n^2}$ eV

Excitation energy $1 \to n$: $13.6\,Z^2\left(1 - \dfrac{1}{n^2}\right)$ eV

H: 1st excitation = 10.2 eV, 2nd = 12.09 eV, ionisation = 13.6 eV
He^+: $E_1 = -54.4$ eV; Li^{2+}: $E_1 = -122.4$ eV

⋆ Common Pitfall

- Excitation potential (V) is numerically equal to excitation energy (eV)
- An atom absorbs only exact energy differences (photon); a colliding electron needs KE ≥ ΔE and keeps the rest

6. HYDROGEN SPECTRUM

$$\frac{1}{\lambda} = RZ^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right), \qquad n_2 > n_1$$

Series	n_1	Region	λ_{max} / λ_{min} (H)
Lyman	1	UV	$\frac{4}{3R}$ (1216 Å) / $\frac{1}{R}$ (912 Å)
Balmer	2	Visible	$\frac{36}{5R}$ (6563 Å) / $\frac{4}{R}$ (3646 Å)
Paschen	3	IR	$\frac{144}{7R}$ / $\frac{9}{R}$
Brackett	4	IR	$\frac{400}{9R}$ / $\frac{16}{R}$
Pfund	5	Far IR	$\frac{900}{11R}$ / $\frac{25}{R}$

First line: $n_2 = n_1 + 1$ (λ_{max}); Series limit: $n_2 = \infty$ (λ_{min})

Number of spectral lines (sample of atoms from level n):

$$N = \frac{n(n-1)}{2} \qquad \text{between } n_1, n_2: \ \frac{\Delta n(\Delta n + 1)}{2}$$

Single atom: maximum $n-1$ lines

⋆ Common Pitfall

- Longest λ = smallest energy jump = first line
- Only Balmer series lies (partly) in the visible region
- Absorption spectrum of H at room temp. shows only Lyman lines (atoms in $n = 1$)

7. RECOIL & REDUCED MASS

Photon momentum $p = \frac{h}{\lambda} = \frac{E}{c}$

Recoil speed of atom: $v = \frac{E}{Mc}$, recoil KE $= \frac{E^2}{2Mc^2}$

Reduced mass: $\mu = \frac{mM}{m+M}$; $E_n \propto \mu$, $r_n \propto \frac{1}{\mu}$

Positronium ($e^- e^+$): $\mu = m/2 \Rightarrow E_1 = -6.8$ eV

⋆ Common Pitfall

- Recoil energy is tiny; normally neglected unless asked
- Muonic atom: $\mu \approx 207\, m_e \Rightarrow$ much smaller orbit, larger energy

8. JEE SHORTCUTS & MASTER TIPS

1. $E_n = -13.6Z^2/n^2$ eV; $r_n = 0.529\, n^2/Z$ Å
2. $K : U : E = 1 : -2 : -1$
3. $\lambda\,[\text{Å}] = 12400/\Delta E\,[\text{eV}]$
4. Lines from level n: $n(n-1)/2$
5. $r_0 = \frac{2kZe^2}{K}$; $b = \frac{r_0}{2}\cot\frac{\theta}{2}$
6. Same λ in different ions: $Z^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$ equal

⋆ Master Confusion List

1. λ_{max} **vs** λ_{min}**:** first line vs series limit
2. **Ionisation energy** from state n is $13.6Z^2/n^2$, not $13.6Z^2$
3. $r \propto n^2$ but $v \propto 1/n$ — don't mix
4. **Sample vs single atom:** $n(n-1)/2$ vs $n-1$ lines
5. α **charge:** $2e$ in closest approach
6. **Energy level spacing** decreases as n increases

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